Ginkgo Bioworks

From Synthetic-Biology Unicorn to AI & Robotics Pivot

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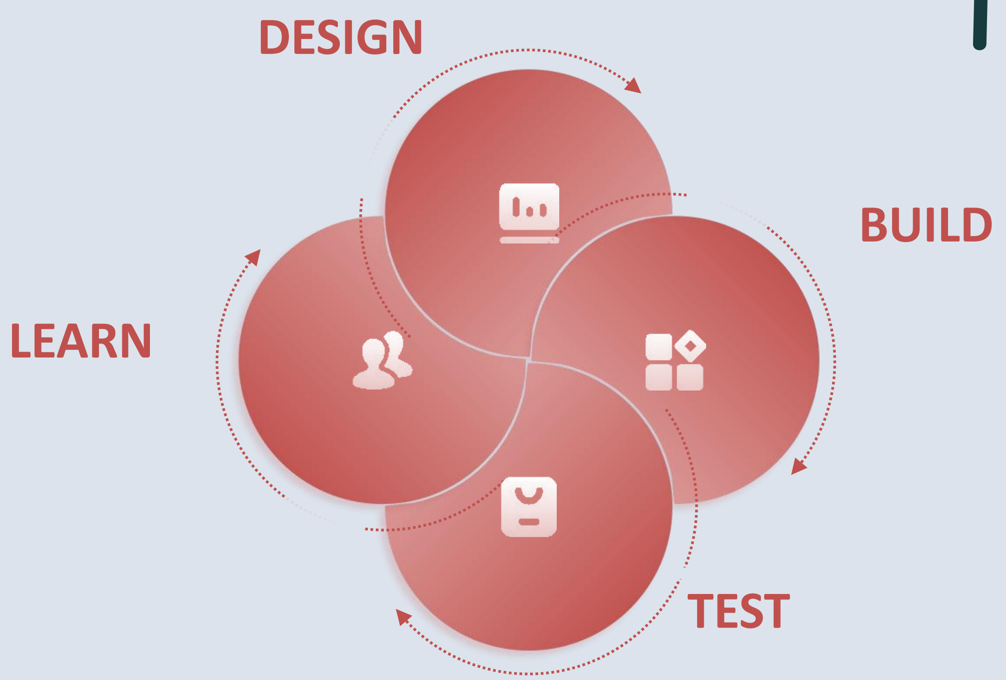
Background

- **Founded 2008, Boston**
  Started in 2008 in Boston by MIT scientists, including CEO Jason Kelly.
- **Goal**
  Make it easy to "program" living cells, almost like writing software.
- **Public via SPAC, 2021**
  Became a public company in 2021 through a SPAC deal; was once worth close to $20 billion.
- **Two business lines**
  Cell Programming (called "the Foundry") and Biosecurity.

Innovative product; The Foundry

- It is a cell programming platform.
- Uses automation, AI and genetic engineering to design and modify microorganisms (bacteria, yeast, fungi).
- In 2026, the system is becoming more AI-powered, aiming to work 5–10 times faster.



The Biosecurity Business

- **Pandemic-era launch**

Grew quickly during COVID-19 to test for and track diseases.

- **Expanded scope**

Later expanded to monitor health threats in places like airports and wastewater.

- **A major revenue stream**

Became a second major source of income but one that depended heavily on pandemic-driven demand.

Applications across Industries

Pharma & Biotech

Helps drug companies to create engineered cells for research

Agriculture

Makes microbes that help crops grow more sustainably

Industrial & Consumer

Produces flavors, scents, and specialty ingredients

Public Health

Tracks disease outbreaks and monitors biosecurity risks

Timeline -Market Entry

2015-2018

First Foundries & Market Expansion(Bioworks4)

2019-2020

Ginkgo Biosecurity & launched COVID-19 testing infrastructure

2021

Public Listing on the New York Stock Exchange under the ticker DNA

2024 – 2025

Shifted to Ai- infrastructure platform & Partnered with hyperscale cloud providers

2026

The Nebula Platform & Cloud Lab

Market Impact

- **Cautionary Tale**

Ginkgo's story is now seen as a warning sign for similar "platform" biotech companies.

- **Technology is not enough**

Shows that having great technology doesn't guarantee a reliable, growing business.

- **Hidden Risk**

Demonstrates how short-term demand spikes (like COVID testing) can hide a company's real long-term health.

- **Industry Shift**

Many biotech companies are now shifting their pitch toward AI and automation to attract investors.

Challenges

01

Revenue Decline & fewer projects

Issue: Revenue dropped significantly due to business restructuring and asset sales.

Impact: Fewer active customers and revenue-generating projects.

02

Persistent Cash Burn & Profitability

Issue: Ginkgo continues to face large financial losses despite reducing costs.

Impact: The company must manage its cash carefully while working toward profitability.

03

Business Model Transition

Issue: Ginkgo is changing its business model to offer automated laboratory services.

Impact: The company needs to attract customers and prove that this new model can generate profit.

Key Takeaways

- Great science does not automatically mean great business
- Relying on a second business line is risky if that line is also unstable.
- Big strategic changes (like selling off a division) can help a company survive, even if it hurts early investors.
- Future biotech founders should plan for both scientific risk and market risk.

THANK YOU

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